2010 Final – Independently Corrected Solutions

Scope. All original Q1–Q8, totaling 130 printed points. The local 12-page answer PDF contains six graded booklets. Grades are comparison evidence, never a substitute for checking the original. Analog results use declared ideal models; no measured process result is claimed. **1. Brief answers**

(a) Use common cell height and aligned V_{DD}/V_{SS} rails for row abutment; use legal width increments and accessible signal pins while obeying the process design rules.

(b) Capacitive external load power and internal-cell switching power (internal nodes), with their individual transition activities. This question says switching power; a separate short-circuit current contribution should not replace its internal-capacitance component.

(c) Balance arrival times at reconvergent inputs; remove hazards with suitable logic factoring/consensus terms or suppress unnecessary transitions before they propagate. An output register limits downstream observation but does not eliminate energy already spent in preceding glitching gates.

(d) Reduce package/supply inductance with short, wide paths and multiple power/ground pins; use local decoupling and control simultaneous-switching current/slew. Bounce follows $L\,di/dt$ plus resistive drop.

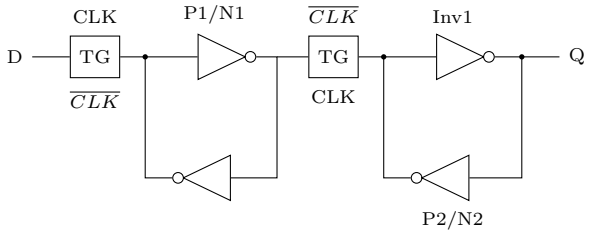
(e) Charge sharing and leakage/noise sensitivity of a floating dynamic node. Direct same-phase dynamic cascades also risk irreversible false evaluation.

(f) Ordinary N-domino implements noninverting cascades; evaluation inputs must obey the monotone rising/stable-input protocol. A dynamic node cannot recharge after a false discharge in the same evaluation phase.

(g) A core-limited die is sized mainly by internal logic/memory area; a pad-limited die is sized mainly by I/O pad perimeter/spacing requirements.

(h) Convert the circuit/netlist into physical geometry: partition/floorplan, place cells, synthesize clock routes, route interconnect and verify physical/timing rules. Physical design is not the act of wafer fabrication.

2. The drawn low-cost register



Upper/lower TG controls are PMOS/NMOS gates.
Both feedback inverters remain connected.

The master TG has n-gate $\overline{CLK}$ / p-gate CLK, so it is transparent low; the slave n-gate CLK / p-gate $\overline{CLK}$ is transparent high. Two data inversions cancel: **a conditional positive-edge, noninverting register**. Always-connected feedback makes each half ratioed; transparency requires the input drive to overpower the old state. Complementary clocks require suitable transition/nonoverlap timing in a physical implementation.

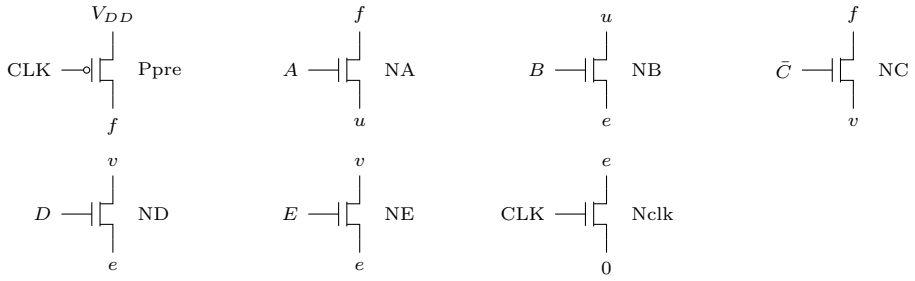
For writing the slave input through its enabled TG, an old-high input is opposed by P2 and driven low by master N1. An old-low input is opposed by N2 and driven high by master P1. At the stated half-supply trip, sufficient divider inequalities are

$$\boxed{R_{P2} > R_{N1} + R_{TG,low}, \qquad R_{N2} > R_{P1} + R_{TG,high}.}$$

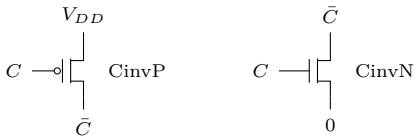
In the explicitly ideal zero-resistance TG limit, these reduce to $R_{P2} > R_{N1}$ and $R_{N2} > R_{P1}$. The comparison is *cross-device*, not pMOS against pMOS and nMOS against nMOS. Resistance scales inversely with each type's width; gate dimensions cannot be made numeric without drive/TG parameters. The master must likewise be writable by D. These inequalities test static write direction, not a specified write time.

3. Three complete transistor implementations Let $H = AB + \bar{C}(D + E)$, so $f = \bar{H}$. In a static network, parentheses are series and $\|$ is parallel. The following drawings explicitly label every physical gate/channel terminal.

Dynamic: seven core MOS. Precharge f high at CLK0; CLK1 enables the footer. The five-transistor PDN implements H. Stable evaluation inputs produce $f = \bar{H}$; the raw dynamic node has no restoring inverter.



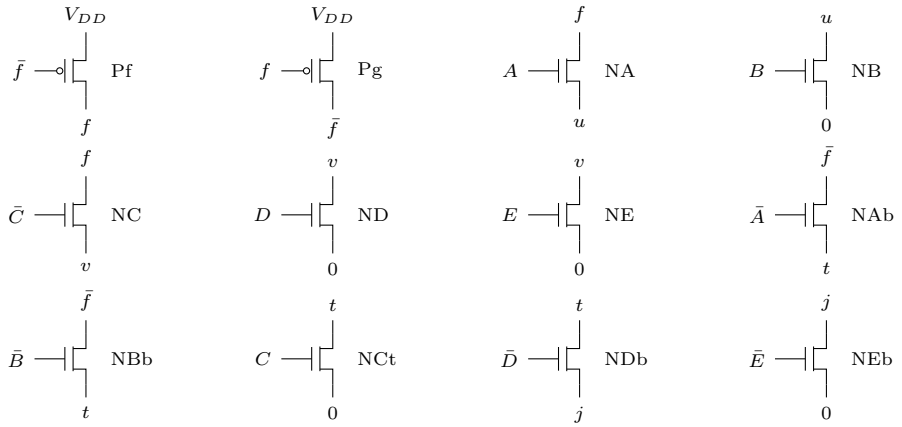
Dynamic core; add the C-complement inverter shown below. Equal terminal names denote connected wires. nMOS bodies use GND, pMOS bodies V_{DD}.



Local complement generation for dynamic/static cores. Equal terminal names denote connected wires. nMOS bodies use GND, pMOS bodies V_{DD}.

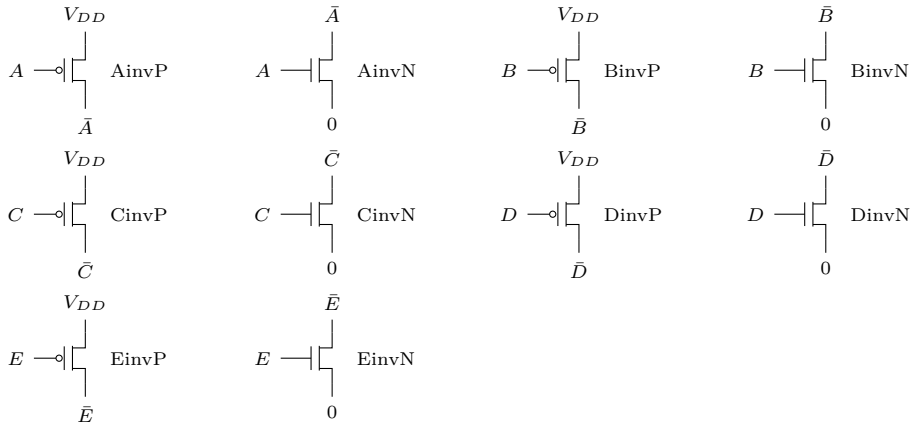
This is 9 MOS including local $\bar{C}$. Hold-state leakage/charge sharing and the monotonic input condition remain relevant.

Fully differential: twelve core MOS. The f PDN implements H; the $\bar{f}$ PDN implements $(\bar{A}+\bar{B})(C+\bar{D}\bar{E}) = \bar{H}$. Cross-coupled pMOS loads restore the opposite output. Exactly one PDN conducts for all 32 input words.



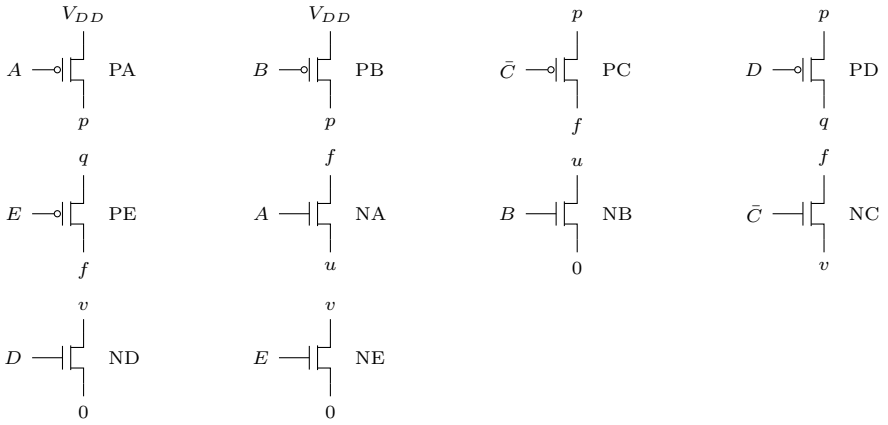
DCVSL core; f is the original required output. Equal terminal names denote connected wires. nMOS bodies use GND, pMOS bodies V_{DD}.

Fully differential complement generators. If only true primary inputs are available, generate all five complements, adding 10 MOS for a total of 22. The core itself assumes these full-rail complementary inputs, not floating gate terminals.



All five complementary inputs. Equal terminal names denote connected wires. nMOS bodies use GND, pMOS bodies V_{DD}.

Static CMOS: ten core MOS, twelve including the C inverter. The PDN is $AB \parallel [\bar{C}(D \parallel E)]$. Its pMOS dual is $(A \parallel B)[\bar{C} \parallel (DE)]$; because pMOS turns on for a low gate, its conduction condition is precisely $\bar{H}$.



Complementary static core; C inverter already shown. No clock. Equal terminal names denote connected wires. nMOS bodies use GND, pMOS bodies V_{DD}.

Do not attach an output inverter to these cores without also changing the required polarity. The saved graph checks all 32 Boolean inputs with exclusive pullup/pulldown conduction, and both dynamic phases.

4. Balanced dynamic sizing and the impossible clock-power target

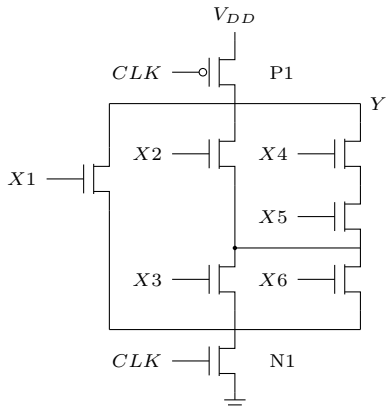


Figure 2: A dynamic logic circuit.

The five minimal discharge paths are X1–N1, X2–X3–N1, X2–X6–N1, X4–X5–X3–N1, X4–X5–X6–N1. The two bottom X3/X6 branches are parallel, not two independent whole trees.

Let R_P be P1's ON resistance. At common $L = .18\,\mu$m and $\mu_n = 2\mu_p$, an nMOS of width W has $R_N/R_P = 2/W$ for W in μm. Matching all minimal paths to R_P matches the *worst* fall time to rise time; parallel simultaneous conduction makes some fall times faster.

One simple valid sizing is

Device	X1	X2	X3	X4	X5	X6	N1
W (μm)	4	8	8	16	16	8	4
R_N/R_P	1/2	1/4	1/4	1/8	1/8	1/4	1/2

All lengths remain $.18\,\mu$m. For a general family, let $q = R_{N1}/R_P$ and $a = R_{X2}/R_P$, with $0 < q < 1$, $0 < a < 1 - q$. Then

$$R_{X1}/R_P = 1 - q, \quad R_{X3}/R_P = R_{X6}/R_P = 1 - q - a, \quad R_{X4}/R_P = R_{X5}/R_P = a/2.$$

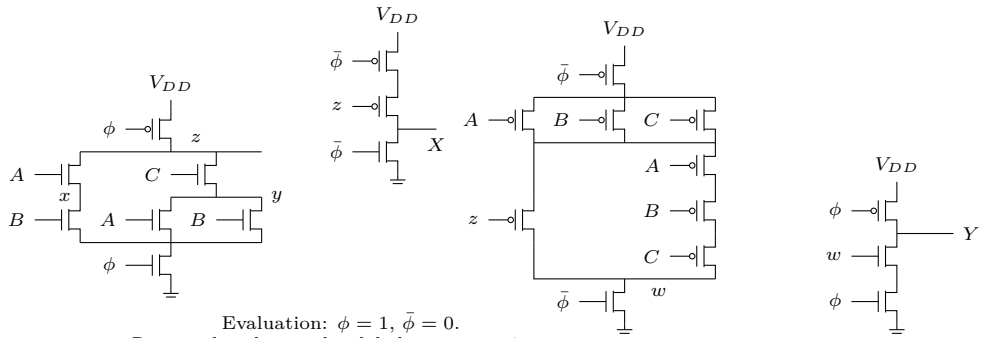
Convert each ratio r to width $W = 2/r$ μm. Internal diffusion loads are excluded by this elementary matching model.

(b) The clock drives P1 and N1. P1 alone has capacitance C; N1 capacitance is $(W_{N1}/4)C$. Therefore

$$P_{clk} = fCV_{DD}^2(1 + W_{N1}/4) < 1.5fCV_{DD}^2 \quad \Longrightarrow \quad W_{N1} < 2\,\mu m.$$

But any finite balanced path must have $R_{N1} < R_P$, which requires $W_{N1} > 2\,\mu m$. **No finite positive sizing satisfies both requirements.** Even the equality limit $W_{N1} = 2$ consumes exactly $1.5fCV_{DD}^2$ and leaves no resistance budget for the functional devices. Scaling P1 would alter a fixed premise. A graded student size cannot override this contradiction.

5. NP timing and correct sharing-only states



Evaluation: $\phi = 1$, $\bar{\phi} = 0$.
Repeated node z and w labels connect wires.

During evaluation, $z = \overline{AB + AC + BC}$, X is majority, and Y is three-input parity. The middle p-dynamic node w rises for 000 or exactly two ones. The z device is below the three parallel pMOS inputs, so it is not a branch directly bypassing them. The final gate is clocked; w drives only its one nMOS input.

The z load is $4C$ (one X p-gate and one middle p-gate), and the w load is C. Output X/Y loads are not supplied and drain capacitance is explicitly ignored. Keep symbolic external loads C_X, C_Y before taking that ideal limit.

Reachable path	Source-defined average delay
inputs $\to z \to X$, 111	$40RC/7 + 3RC_X/2$
inputs $\to z \to X$, exactly two ones	$8RC + 3RC_X/2$
primary inputs $\to w \to Y$, 000	$8RC/3 + 3RC_Y/2$
inputs $\to z \to w \to Y$, exactly two ones	$10RC + 3RC_Y/2$

For z discharge, 111 gives $R + (2R \parallel 1.5R) = 13R/7$; two ones give $3R$. Its precharge resistance is R. For w, 000 gives $R + R/3 + 3R = 13R/3$, while two ones give $3R$; reset resistance is R. Thus w's mean delay is $8RC/3$ or $2RC$. A z-driven w transition requires two ones, so $8RC$ and $2RC$ sensitize together. Combining independent extrema from incompatible words does not establish a physical path. Under the printed mean-delay metric and $C_X = C_Y = 0$,

$$\boxed{t_{cd} = 8RC/3, \qquad t_{pd} = 10RC.}$$

This is the source's averaged delay abstraction, not a measured minimum/maximum transient bound. The zero final-stage RC is the unloaded/parasitic-neglected limit, not instantaneous physical switching. Evaluation inputs must settle/obey monotonicity; arbitrary input races are outside this estimate.

(b) A sharing-only case must have no actual ground path. With z precharged to 1.8 V and internal nodes initially zero, 001 connects only y to z:

$$V_z = 1.8\frac{2}{2 + .8} = \frac{9}{7}\,V \simeq 1.2857V.$$

For 100, only x joins z, giving 1.5 V. Both exceed the pMOS turn-on boundary $1.8 - .6 = 1.2$ V; 010 does not connect either internal capacitor to z. Therefore **the specified ideal sharing test is safe**, with a small 1/7-V worst margin to pMOS enable. Pooling both x and y is invalid: A=C=1 opens a real majority discharge path. This conclusion uses the 2010 values; another year's smaller C_z or different threshold would change the test. Leakage, charge injection, finite evaluation time and noise are not quantified by this test.

6. Signed skew, setup and hold Define $\delta = t_{capture} - t_{launch}$. Setup at $T = 10$ ns requires

$$t_{pd,logic} \leq 10 - .8 - 1.6 + \delta = 7.6 + \delta.$$

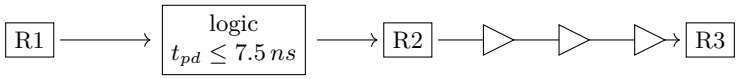
The printed .1 ns does not give a sign. For a skew magnitude bounded by .1 ns, the guarantee is $\boxed{t_{pd,logic} \leq 7.5ns}$; capture late by .1 ns would allow 7.7 ns. Hold is

$$.4 + t_{cd,logic} \geq 1.2 + \delta.$$

On the direct R2–R3 path, assuming the printed .2/.3 ns wire/buffer delays are also their minimum delays, n noninverting buffers give

$$.4 + .2 + .3n \geq 1.2 + .1 \quad \Longrightarrow \quad n \geq 3.$$

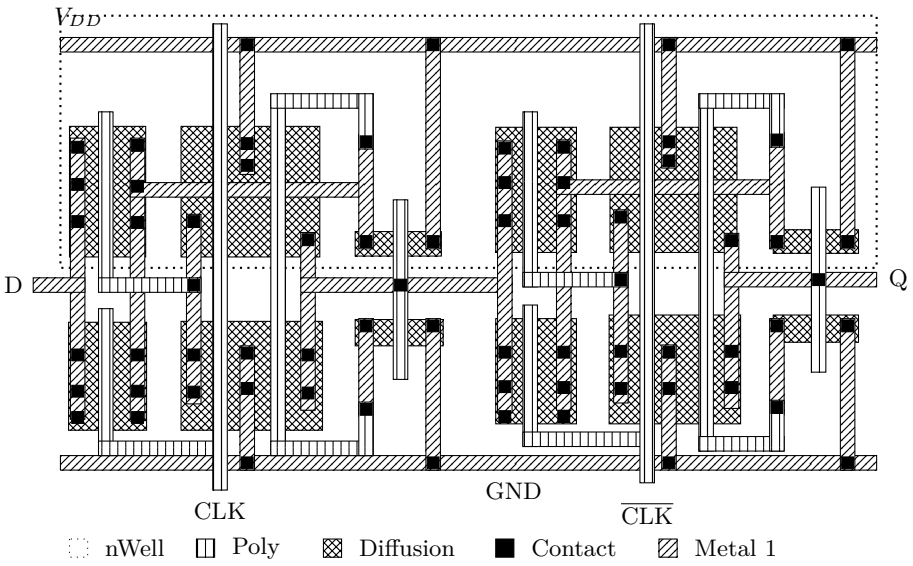
Two suffice only for nominal zero/early capture skew. Three buffers guarantee the stated magnitude case and give setup requirement $.8 + .2 + .9 + 1.6 = 3.5ns$, safely below 9.9 ns. Retain R1–R2's7.5 ns logic bound.



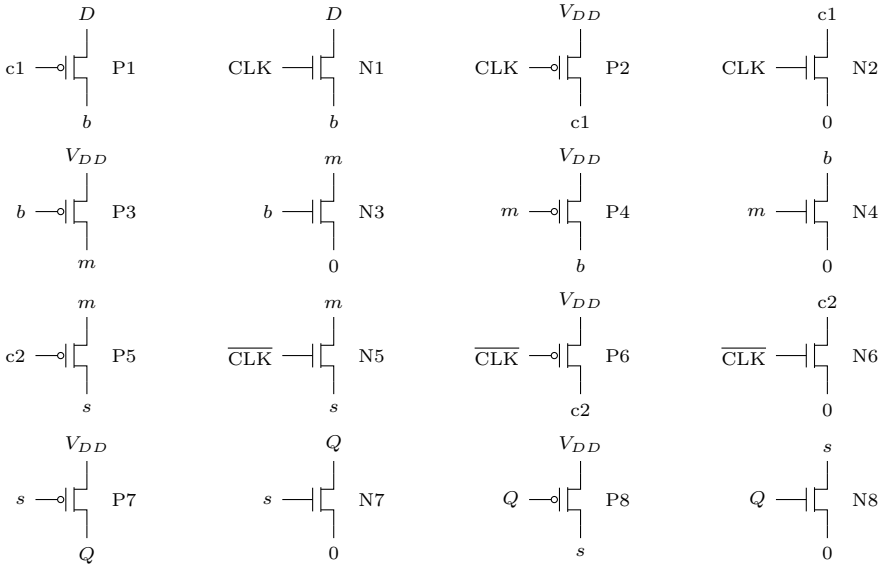
3 noninverting buffers, .3 ns each; wire .2 ns total

All three registers retain the original common clock; the added buffers are on the data path. For a physical guarantee, use actual contamination delays: $t_{ccq} + t_{cd,wire} + \sum t_{cd,buf} \geq t_{hold} + \delta$. Propagation delay alone does not certify hold. The numerical three-buffer answer uses the explicitly declared single-delay classroom convention.

7. Literal sixteen-MOS layout extraction



Geometry normalized; all eight physical gate columns and contacts retained.



Eight physical columns, 16 MOS; all local clock inverters and always-connected feedback preserved. Equal terminal names denote connected wires. nMOS bodies use GND, pMOS bodies V_{DD}.

The first device pair is a TG whose channel starts at D. Master: TG D↔b enabled at CLK=1; clock inverter CLK→c1; storage inverter b→m; fixed feedback inverter m→b. Slave: TG m↔s enabled at CLK=0; clock inverter CLKB→c2; storage inverter s→Q; fixed feedback inverter Q→s. The 16 physical MOS count follows four crossings per row per half. No outer input/output inverters exist in this original.

With sufficient write drive, the master follows while CLK is high and holds at the falling edge; the slave then passes that stored word. Two data inversions cancel, so the conditional circuit is a **negative-edge, noninverting register**. Its clock polarity differs from the positive-edge Figure 1 in Q2. Keep each source's clock routing. Feedback remains connected during writing and can oppose new data, so the original without sizes does not prove writeability or settling. Replacing it with a two-TG latch would change its electrical connections.

8. Upper Schmitt trip: use the given voltage and bias-dependent currents

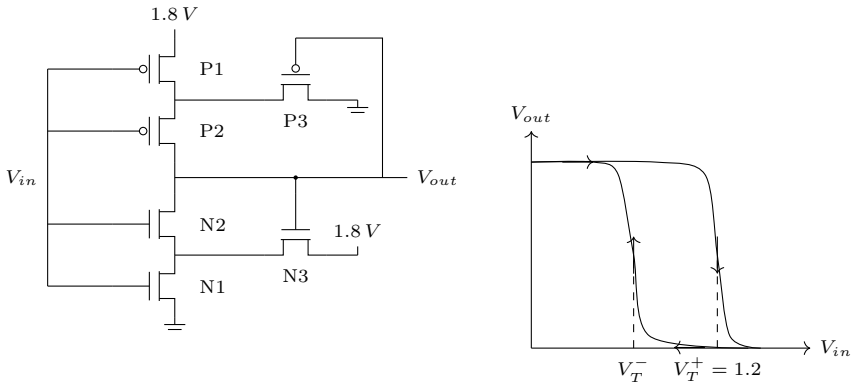


Figure 6: Schmitt circuit and qualitative VTC.

The original explicitly states $V_T^+ = 1.2$ V. Just before the rising trip, $V_{out} \simeq V_{DD}$ keeps N3 on. N2 begins conduction when internal node a satisfies $V_a = V_T^+ - V_{tn} = .6$ V. Neglect body effect/channel-length modulation and use the standard long-channel model. N1 is at the saturation boundary with overdrive .6 V; N3 has gate 1.8 V, source .6 V, drain 1.8 V and saturation overdrive .6 V. Current balance is

$$\frac{\beta_{N1}}{2}(1.2 - .6)^2 = \frac{\beta_{N3}}{2}(1.8 - 1.2)^2, \qquad \boxed{\beta_{N1}/\beta_{N3} = 1}.$$

Treating both ON resistances as bias independent and inverting a divider ratio can instead produce 2; that is not a valid beta extraction. The devices have different gate/source biases, so resistance cannot be inferred as a common constant times the inverse of beta. At this onset, their equal currents coexist with $V_{DS,N1} = .6$ V and $V_{DS,N3} = 1.2$ V: therefore $R_{N3} = 2R_{N1}$ is consistent with *equal beta*. The different bias coefficients explain why a resistance ratio of two does not imply a beta ratio of two. A full switching point including the pMOS network/body effect would need extra parameters; the expression above is the threshold-onset sizing approximation with its assumptions visible.

Sources and checks. The original Fall 2010 prompt governs every premise. All 12 student PDF pages were compared locally; identity fields are not redistributed. Saved standard-library graph checks cover 32 logic words, two dynamic phases, 64 dynamic-gate inputs, 16 literal-layout settled/contention states, and exact path/clock/share/skew constraints. Reference mechanisms are checked against the instructor's CMOS electrical characteristics, advanced logic and sequential circuits. No grader's score is correctness proof.